

ME381 ROBOTICS

Experiment No.: 9

Experiment Title: Mobile robot control and its application

Group No.: B6

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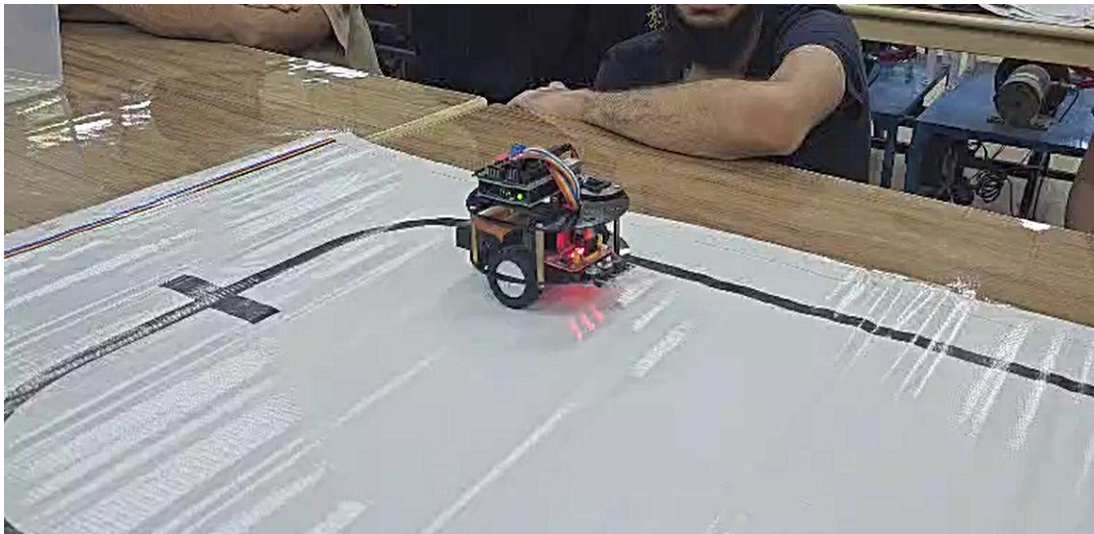
Roll no.: 230793

Results for the group task:

Experiment 1: Experiment on Motion of an Open Loop Mobile Robot

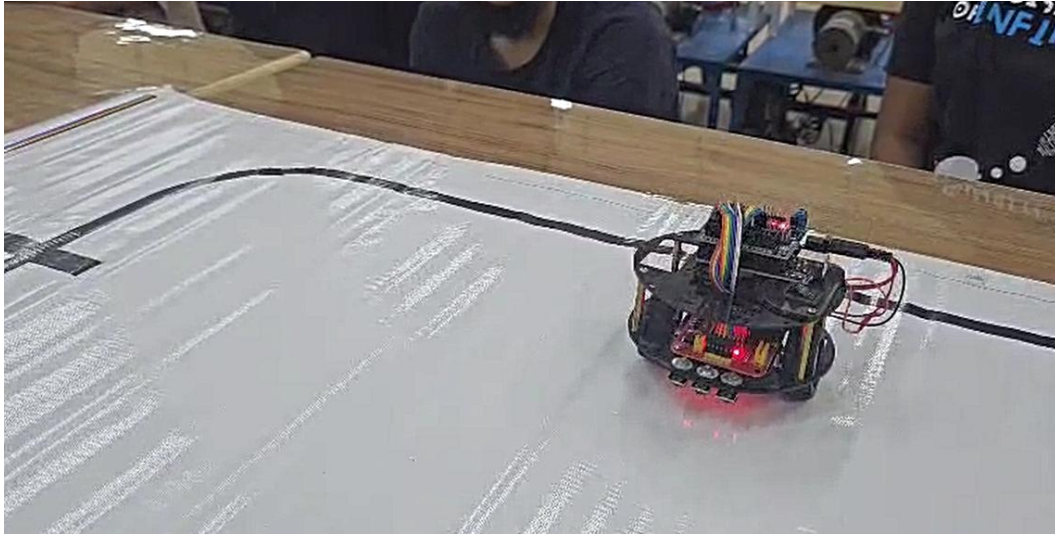
Part 2: Navigating a Rectangular Path

Mobile robot is at 1st corner

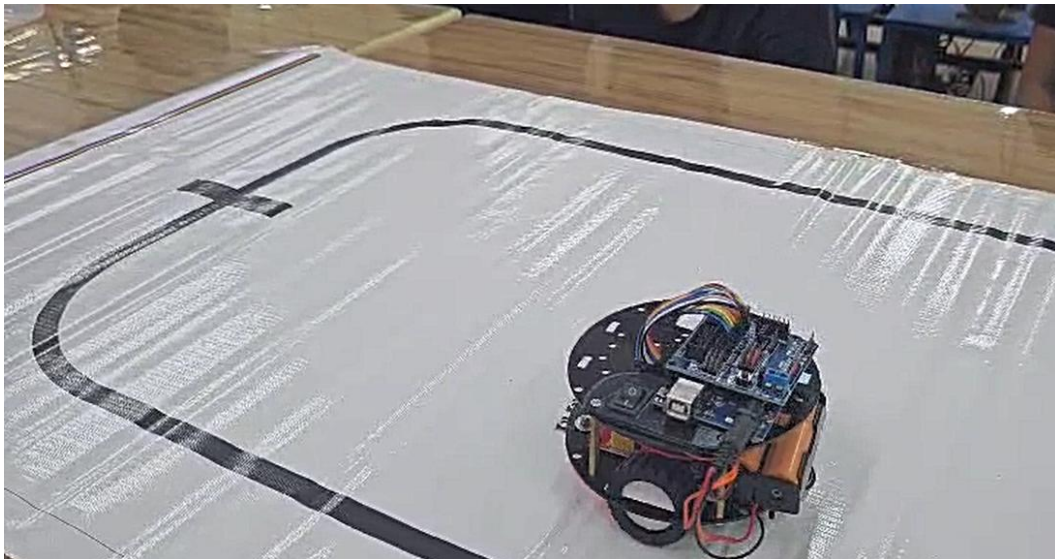


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Mobile robot is at 2nd corner

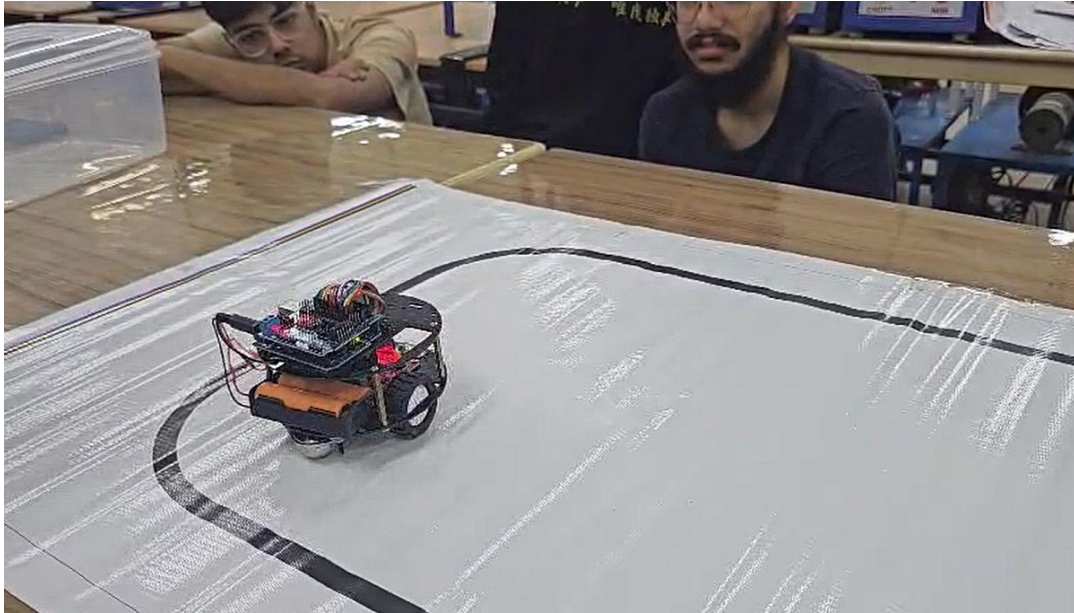


Mobile robot is at 3rd corner

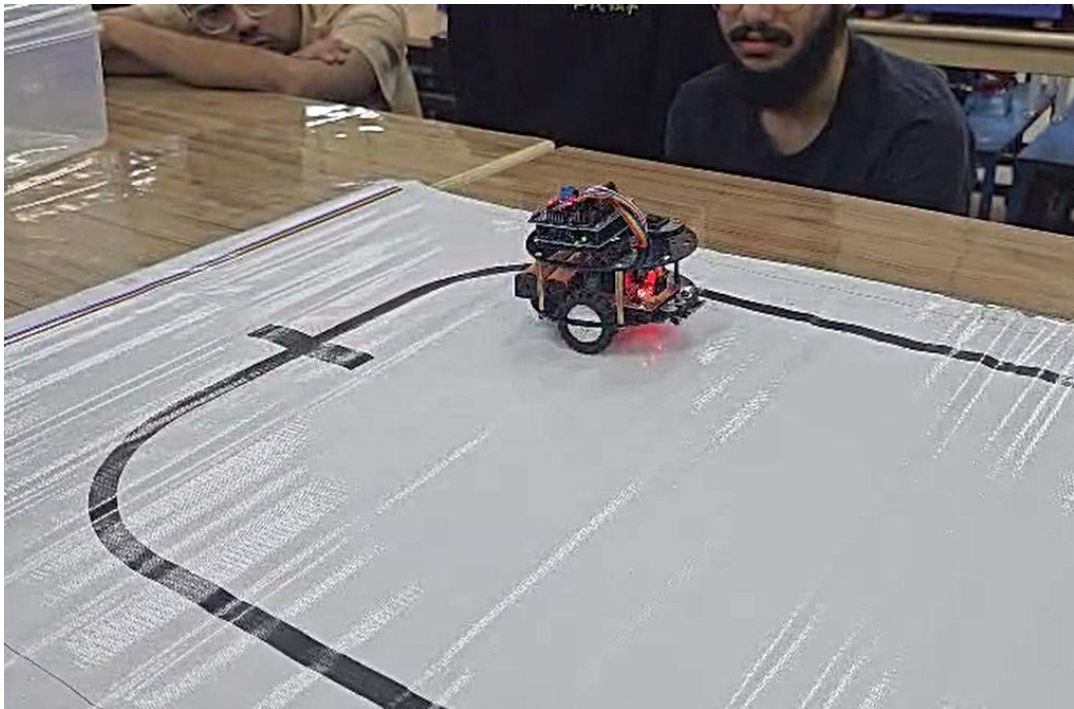


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Mobile robot is at 4th corner



Mobile robot is again at 1st corner



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The time it takes to reach each corner using a stopwatch:

1st to 2nd → 3.5 sec

2nd to 3rd → 2.5 sec

3rd to 4th → 3.5 sec

4th to 1st → 2.5 sec

```
void loop() {  
  Forward();  
  delay(3500);  
  Left();  
  delay(1800);  
  Forward();  
  delay(2500);  
  Left();  
  delay(1800);  
  Forward();  
  delay(3500);  
  Left();  
  delay(1800);  
  Forward();  
  delay(2500);  
  Left();  
  delay(1800);  
}
```

[Link of the video footage of robot's behaviour](#)

Experiment 2. Laboratory Experiment on Obstacle Avoidance in a Mobile Robot

Part 2: Student Observation and Experimentation

How the robot detects obstacles (the presence of students moving in front of it)?

The robot uses **ultrasonic sensors** that emit sound waves and measure the time taken for the echo to return. When a student moves in front, the reflected signal indicates an obstacle within a set distance.

How the robot alters its motion to avoid obstacles (stopping, turning, etc.)?

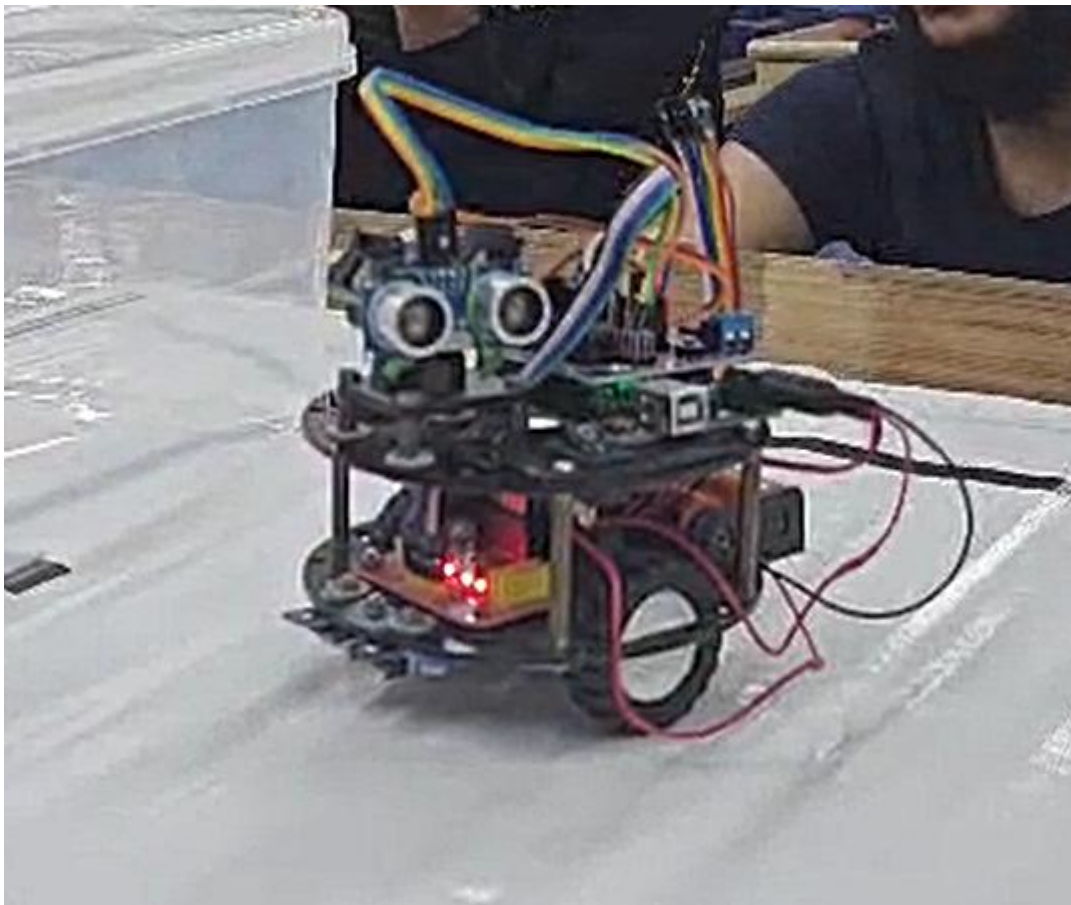
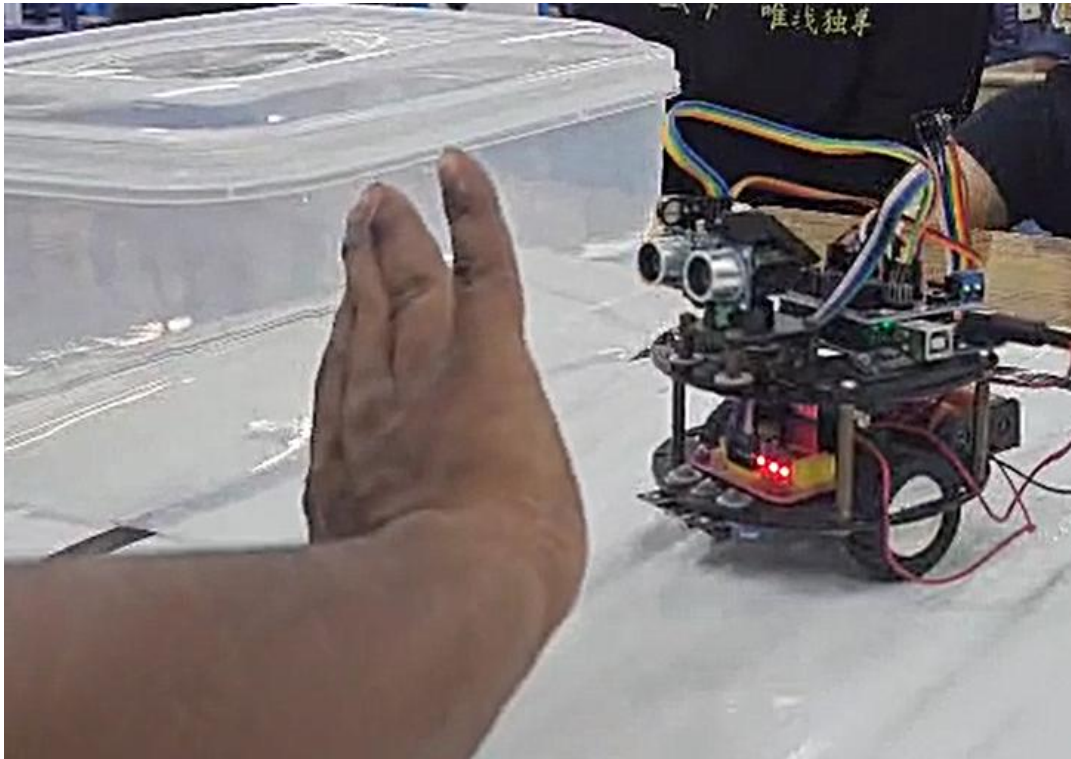
On detecting an obstacle, the robot **stops**, then **turns** left or right depending on which direction is clear, and continues moving forward.

Whether the robot successfully avoids the students and continues moving?

Yes, the robot **successfully avoids the students**, navigating around them and resuming its path once the obstacle is cleared.

[Link of the video footage of robot's behaviour](#)

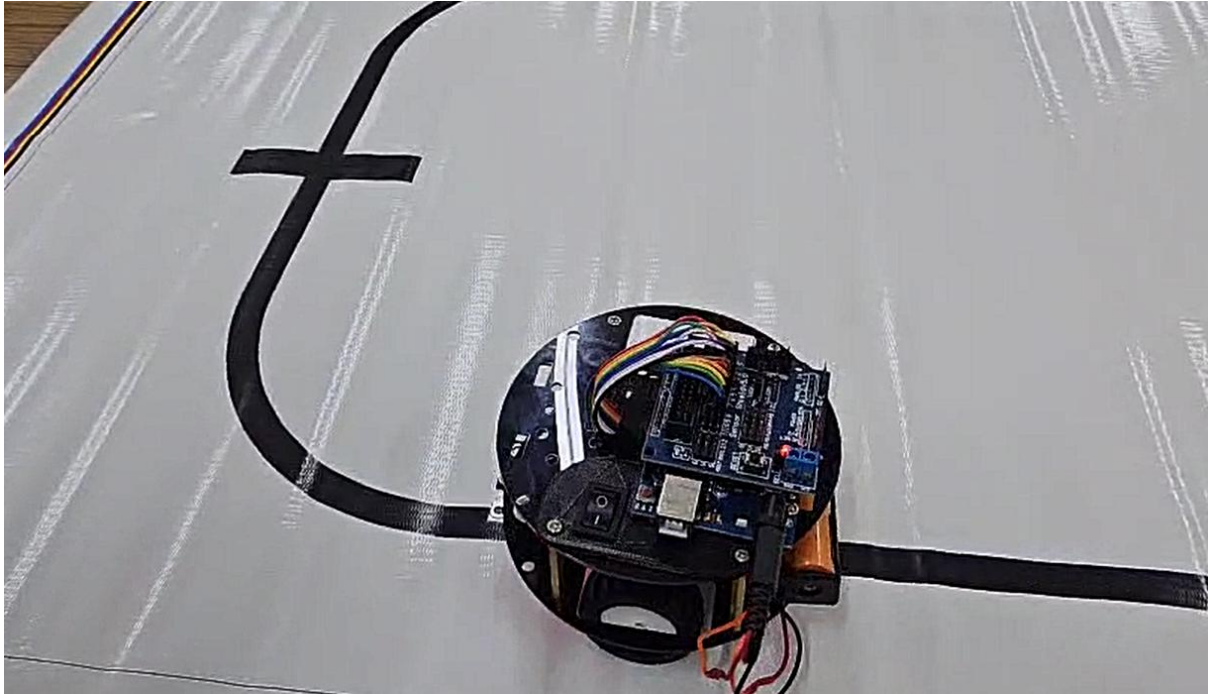
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Experiment 3. Laboratory Experiment on Line Following using IR Sensors

Part 2: Navigating a Rectangular Path



[Link of the video footage of robot's behaviour](#)

Code to handle navigation with two IR sensors:

```
void loop() {  
  // put your main code here, to run repeatedly:  
  Left_Tra_Value = digitalRead(LEFT_LINE_TRACING);  
  Right_Tra_Value = digitalRead(right_LINE_TRACING);  
  
  Serial.println("l,r:");  
  Serial.println(Left_Tra_Value);  
  Serial.println(Right_Tra_Value);  
  
  if (Left_Tra_Value == 1 && Right_Tra_Value == 1)  
  {  
    Stop();  
  }  
  if (Left_Tra_Value == 1 && Right_Tra_Value == 0)  
  {  
    Left();  
  }  
  if (Left_Tra_Value == 0 && Right_Tra_Value == 1)  
  {  
    Right();  
  }  
  if (Left_Tra_Value == 0 && Right_Tra_Value == 0)  
  {  
    Forward();  
  }  
}
```